

Abstract

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The efficacy of vitamin D-fortified dairy products in improving serum 25(OH)D concentration: a systematic review and meta-analysis of randomised controlled trials

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The National Dietary and Nutrition Survey⁽¹⁾ confirmed the high prevalence of suboptimal vitamin D intake and status in the United Kingdom (UK) population. The UK Scientific Advisory Committee on Nutrition report in 2024⁽²⁾ focused on identifying new food vehicles for vitamin D fortification. Given that dairy products are staple foods in the UK diet which may be possible vehicles for fortification, this study aims to examine the efficacy of vitamin D-fortified dairy products for improving serum 25-hydroxyvitamin D [25(OH)D] concentration using data from randomised controlled trials (RCTs).

The search was conducted in PubMed, Embase, and Web of Science. After abstract/title and full-text screen, 35 eligible RCTs with interventions using vitamin D3-fortified cheese ($n = 5$), vitamin D3-fortified milk/milk powder ($n = 15$), vitamin D3-fortified yoghurt/yoghurt drinks ($n = 11$), vitamin D2-fortified milk/milk powder ($n = 3$), and vitamin D-fortified (form unknown) milk/milk powder ($n = 8$) and vitamin D-fortified (form unknown) yoghurt/yoghurt drinks ($n = 4$) were identified and included in the meta-analysis. Meta-analysis of the mean difference (MD) of serum 25(OH)D concentration between the baseline and endpoint was performed in a random effects model using STATA 18.0.

Results showed that serum 25(OH)D concentrations were significantly increased by vitamin D3-fortified milk/milk powder (MD: 17.20 nmol/L, 95% CI: 11.74 – 22.65 nmol/L, $p < 0.001$, $I^2 = 95.96\%$), vitamin D3-fortified yoghurt/yoghurt drinks (MD: 26.22 nmol/L, 95% CI: 18.68 – 33.77 nmol/L, $p < 0.001$, $I^2 = 97.21\%$), vitamin D2-fortified milk/milk powder (MD: 11.61 nmol/L, 95% CI: 9.31 – 13.91 nmol/L, $p < 0.001$, $I^2 = 0\%$), vitamin D-fortified (form unknown) milk/milk powder (MD: 13.58 nmol/L, 95% CI: 8.53 – 18.63, $p < 0.001$, $I^2 = 97.52\%$) and vitamin D-fortified (form unknown) yoghurt/yoghurt drinks (MD: 27.74 nmol/L, 95% CI: 16.83 – 38.64 nmol/L, $p < 0.001$, $I^2 = 91.16\%$). Although vitamin D3-fortified cheese did not show statistically significant results (MD: 16.78 nmol/L, 95% CI: -3.61 – 37.16, $p = 0.11$, $I^2 = 99.33\%$), the results became significant by omitting one study when a leave-one-out analysis was conducted (MD: 24.13 nmol/L, 95% CI: 4.69 – 43.58, $p = 0.015$, $I^2 = 89.89\%$). Subgroup analysis of each dairy product was conducted but heterogeneity remained high in general ($I^2 \geq 75\%$)⁽³⁾.

In conclusion, these findings demonstrate that vitamin D-fortified dairy products are effective in improving serum 25(OH)D concentration.

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References

1. National Dietary and Nutrition Survey (2020) National Diet and Nutrition Survey Rolling programme, Years 9 to 11 (2016/2017 to 2018/2019)
2. Scientific Advisory Committee on Nutrition (2024) Fortifying foods and drinks with vitamin D: main report. [Available at <https://www.gov.uk/government/publications/fortifying-food-and-drink-with-vitamin-d-a-sacn-rapid-review/fortifying-foods-and-drinks-with-vitamin-d-main-report#background>]
3. Higgins JP, Thompson SG, Deeks JJ, Altman DG (2003) BMJ 327, 557-560